

Finite-sample learning dynamics of discrimination recovery in amortized neural IRT

representation sharing throttles discrimination; decoupling is a convergence-rate fix

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Setup: amortized neural IRT

- Encoder (LSTM): response history $\rightarrow$ time-varying ability θ_t
- Item embeddings $\rightarrow$ discrimination $\alpha_i > 0$, difficulty β_i ; GPCM decoder

$$P(Y_i = k \mid \theta) = \frac{\exp \sum_{m=1}^k \alpha_i (\theta - \beta_{i,m})}{\sum_{l=0}^{K-1} \exp \sum_{m=1}^l \alpha_i (\theta - \beta_{i,m})} \quad (2\text{PL at } K=2)$$

- Trained on response prediction only. **No supervision** on θ, α, β , no priors.
- Scale is gauge-free ($\theta \rightarrow s\theta + t, \alpha \rightarrow \alpha/s$) $\Rightarrow$ recovery = rank (Spearman ρ).

Architecture: shared vs decoupled item code

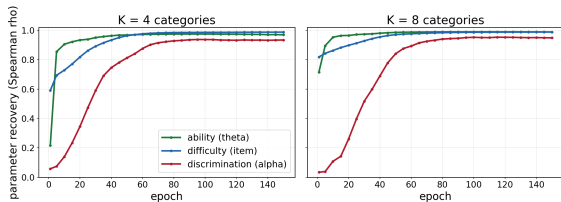
The item code enters twice: the encoder ($\rightarrow \theta$) and the α, β readouts.

- **Shared:** one table, width W (`item_key_dim=None`)
- **Decoupled:** `emb_dim=8` + separate `item_key_dim=W-8` feeding only α, β
- `state_alpha=True` in both $\Rightarrow$ the study varies the *representation*, not the readout

$$\text{shared: } \theta \leftarrow e_q \rightarrow (\alpha, \beta) \quad \text{decoupled: } \theta \leftarrow e_q^{\text{val}}, \quad e_q^{\text{key}} \rightarrow (\alpha, \beta)$$

Discrimination is the low-information parameter

Ability and difficulty are learned fast; discrimination lags



Gradients ($r = p - y$):

$$\partial_{\theta} L = r\alpha, \quad \partial_{\beta} L = -r\alpha, \quad \partial_{\alpha} L = r(\theta - \beta)$$

Fisher ($w = p(1-p)$):

$$I(\theta) = I(\beta) = \alpha^2 w, \quad I(\alpha) = (\theta - \beta)^2 w$$

- $I(\alpha) \rightarrow 0$ at $\theta = \beta$ (where responses concentrate)
- stiffness $\kappa = I(\theta)/I(\alpha)$ grows with K

Why is discrimination degraded under sharing?

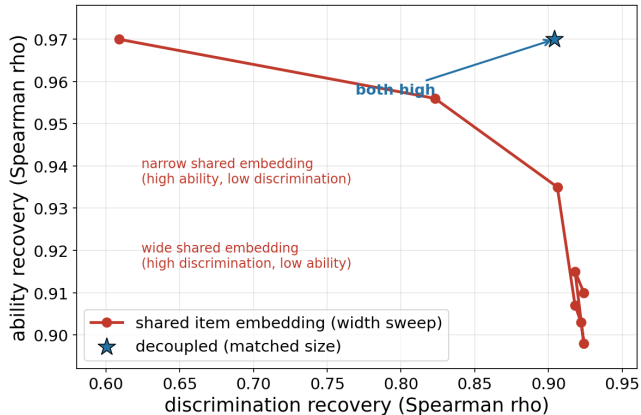
Three explanations, one test each.

- ① **Capacity** → a wider shared code would fix it
- ② **Expressivity** → no shared code holds both directions at the optimum
- ③ **Dynamics** → gradient flow drives the shared code to the θ compromise

Synthetic GPCM, known θ, α, β , LSTM. Each finding eliminates one.

Test 1, not capacity: allocation

One shared embedding cannot get both
decoupling sits above the curve

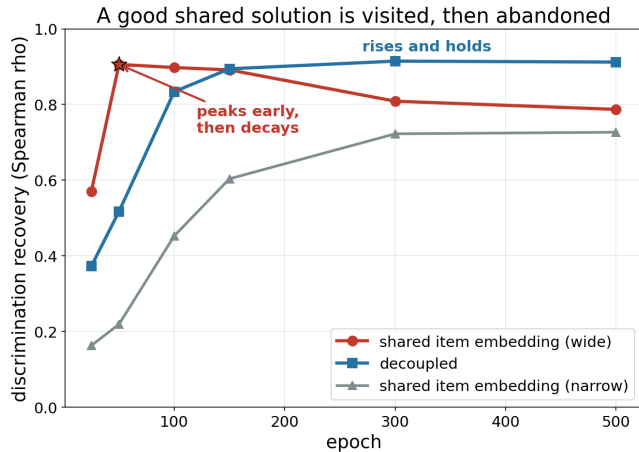


Control: matched total
item-embedding params.

- shared width sweep $\rightarrow$ Pareto frontier, no W gets both
- decoupled sits **above**: $\theta \approx 0.97$, α 0.90–0.94, fewer params

$\Rightarrow$ allocation, not budget.

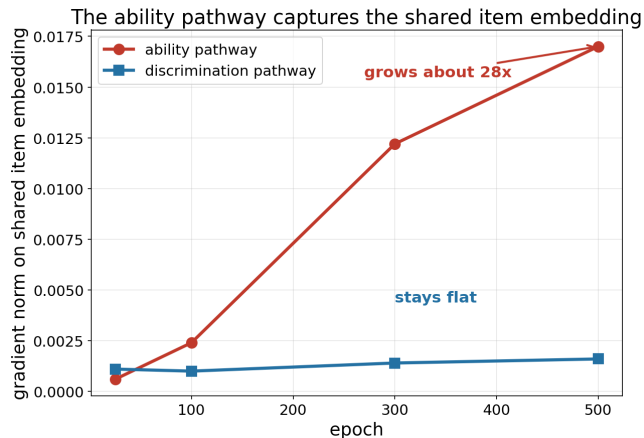
Test 2, not expressivity: reached then left



- wide-shared α : peaks 0.91 (ep 50), decays to 0.79 (ep 500)
- decoupled: rises 0.91, **holds**

⇒ the both-high optimum is reachable, reached, then abandoned = dynamics.

Test 2, mechanism: the shared code is conditioned



- θ -pathway pull grows $\sim 28\times$;
 α -pathway flat
- $\cos(g_\theta, g_\alpha) \approx 0$ (not a tug-of-war)
- the lag is eigenvalue spread,
 $\kappa \approx 2.2$ at $K=4$

fast θ mode first, slow α mode last.

The dynamics as a gradient-flow ODE

Item-code block near the optimum, $\delta = e_q - e_q^*$:

$$\dot{\delta} = -H \delta, \quad \delta_c(t) = \delta_c(0) e^{-\lambda_c t}.$$

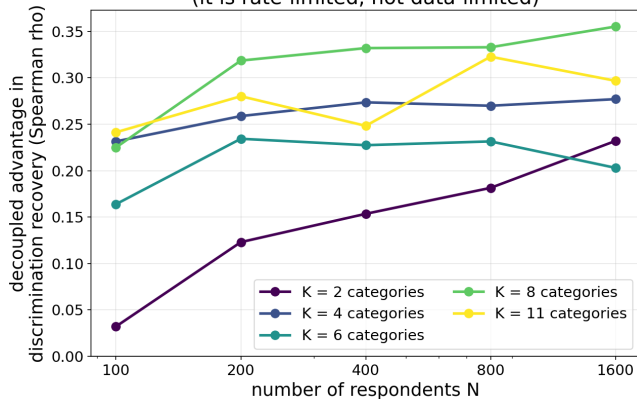
- Shared block, two curvatures: $\kappa = \frac{\lambda_\theta}{\lambda_\alpha} = \frac{I(\theta)}{I(\alpha)} \gg 1$
- One step size $\eta < 2/\lambda_{\max}$: slow mode contracts at $\eta\lambda_\alpha \sim 1/\kappa \Rightarrow \tau_\alpha^{\text{sh}} \sim O(\kappa)$
- Decoupled block: $O(1)$ -conditioned $\Rightarrow \tau_\alpha^{\text{dec}} \sim O(1)$

$\text{speedup} \sim \kappa = I(\theta)/I(\alpha) \text{ (grows with } K\text{)}$

Same fixed point (free-table invariant) $\Rightarrow$ **rate, not endpoint**. Largest under plain GD;
Adam's preconditioner compresses κ .

Test 3, finite-sample: rate-limited not data-limited

More data does not shrink the advantage
(it is rate-limited, not data-limited)

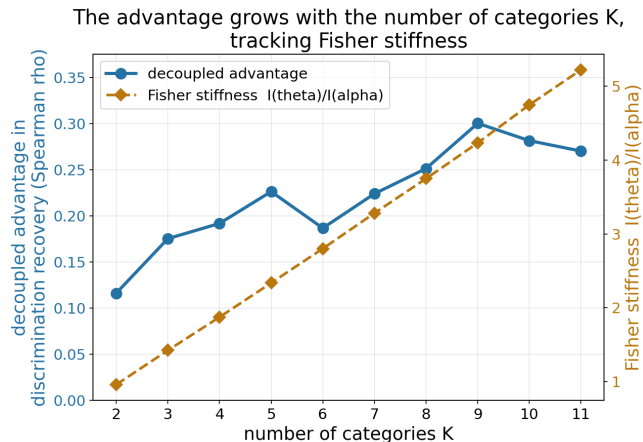


Fixed budget, vary N .

- advantage does not shrink with data ($K=2$: $+0.03 \rightarrow +0.23$)
- more learners, same steps $\rightarrow$ slow α mode further behind

$\Rightarrow$ the finite-sample signature of a rate effect.

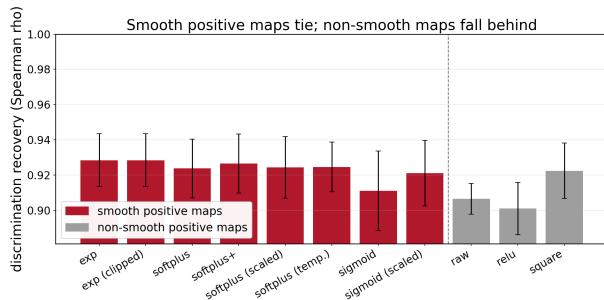
Test 3, the advantage tracks κ



- $K=2$ to 11: Δ_α vs stiffness κ , $\rho = 0.89$
- decoupled de-noises α (σ 0.005–0.042)
- rise then saturate (decoupled ceilings ≈ 0.95)

$\Rightarrow$ size of the effect = the Fisher stiffness, set by K .

Ruled out: the positivity map



$\alpha = g(\cdot)$, g positive. Is $g = \exp$ special?

- matched init + per-map LR: exp ties all smooth maps ($\Delta \approx 0$)
- direct- α control (θ, β frozen): scalar preconditioner does nothing ($\kappa=1$)

$\Rightarrow$ positivity needed, exp not special.
The lever is the representation.

Conclusion

Decoupling the discrimination code buys an $O(\kappa)$ convergence-rate advantage on discrimination rank, $\kappa = I(\theta)/I(\alpha)$, growing with K , no endpoint cost.

- not capacity (matched params), not expressivity (reached then left), but **dynamics**
- a measurement-validity condition: recovered α is otherwise a finite-budget allocation artifact
- positivity map ruled out as the lever

Scope: synthetic GPCM, LSTM (architecture-independent on swap), 3–10 seeds. Open: θ /threshold channels, NRM slopes, GRM, real-data stability.